- 1
  - i
    - $$V_c(t) = V(1 - e^{\frac{-t}{RC}})$$
    - $$V_c(RC) = V(1 - e^{\frac{-RC}{RC}})$$
    - $$V_c(RC) = V(1 - e^1)$$
    - $$V_c(RC) \approx 0.63V$$
  - ii
    - $$|\frac{V_c}{V_{in}}| = \frac{1}{\sqrt{1 + (2\pi fRC)^2}}$$
    - Using $F_c = \frac{1}{2\pi RC}$
    - $$|\frac{V_c}{V_{in}}| = \frac{1}{\sqrt{1 + 1^2}}$$
    - $$|\frac{V_c}{V_{in}}| = \frac{1}{\sqrt{2}}$$
- [[2]]
  -
  - By measuring using the 'cursor' tool on the oscilloscope, we can measured $\Delta t = 440\mu s$.
  - Aka the time taken to reach 63% of the amplitude takes $440\mu s$.
  - I will assume a $\pm 20\mu s$ uncertainty, this is due to the oscilloscope only measuring values in jumps of $40\mu s$. I choose half the differences between possible measurement values.
  - According to part 1, the point in which 63% of the amplitude is reached is at $t = RC$
  - We measured the resistance to be 4.6, with the multimeter. Below is the calculation without error propogation.
  - $$RC = 4.6k\Omega \times 0.1F = 460\Omega f^-1$$
  - Using python.our resistance measurement was taken with a multimeter which has a 1.2% uncertainty. The capacitor specifies only its tolerance and does not have an uncertainty, so a 1% uncertainty has been assumed.
  - ```
    R = ufloat(4600, 4600 * 0.012)
    C = ufloat(0.1e-6, 0.1e-6 * 0.01)
    TCalculated = R * C
    print(TCalculated)
    ```
    ```

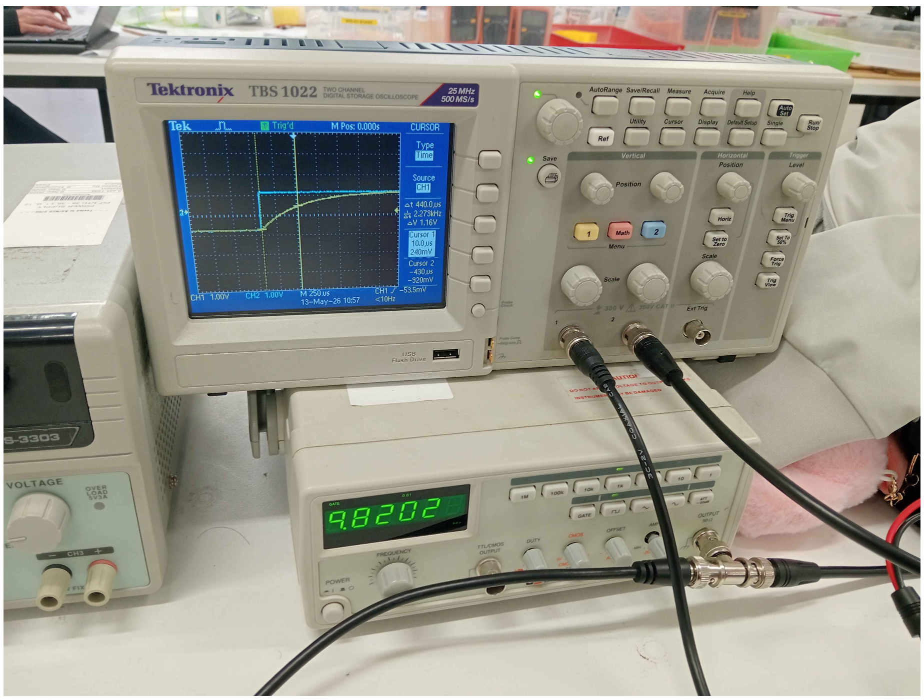

Figure 1: file:///home/burrs/Repositories/School/src/10week/physics/IMG20260513141020.jpg

- The return value is (4.60+/-0.07)e-04. *note: The uncertainties library has been imported and is used for all further python calculations
- Now we have both the measured and calculated values of the milliseconds, we can extrapolate a σ uncertainty for both the values.
-

$$\sigma = \frac{|t_{measured} - t_{calculated}|}{\sqrt{\sigma_1^2 + \sigma_2^2}}$$

- again, using python, and our measured 440 nanoseconds, we can calculate the sigma.
- ```
  TMeasured = ufloat(440e-9, 20e-9)
  Sigma = (TCalculated.n - TMeasured.n) / (TCalculated + TMeasured).s
  print(Sigma)
  ```
- the return value for this is `0.9411057177286997`
- Since this is < 3, we can safely say that our two values corroborate eachother.

• 3

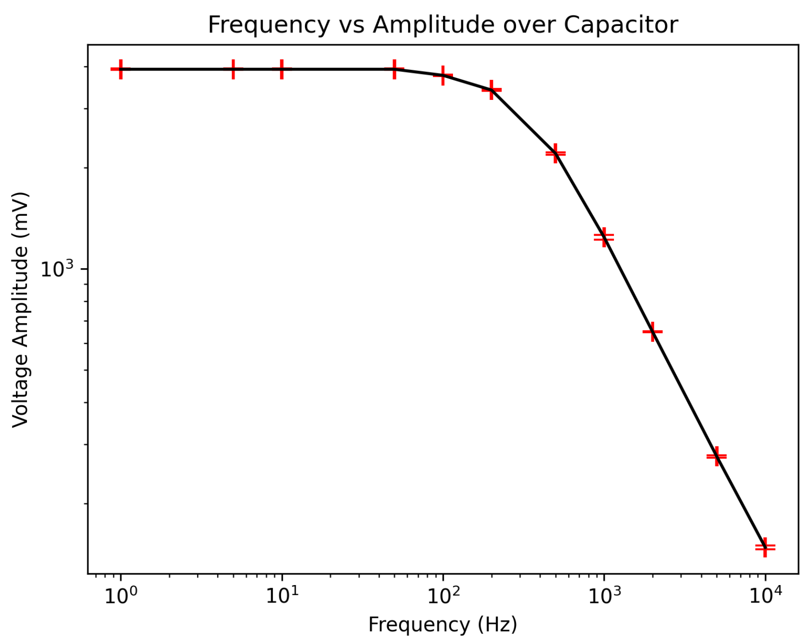

Figure 2: file:///home/burrs/Repositories/School/src/10week/physics/FreqAmpCap.png

-
- Frequency uncertainty is 1%, and voltage amplitude uncertainty is half of the lowest discrete jump. So if the oscilloscope measures 1140mV, the jump between each value is 20mV.
  ```

- I'm going to interpolate and say the cutoff frequency is a point on 10^2 Hz. I will assume a

  $$\sqrt[[May13th,2026]{]} \times 10^2 Hz$$

  uncertainty, just due to how much uncertainty our log graph gives us.
- Calculating Fc in python using $F_c = \frac{1}{2\pi RC}$:
- `print(1/(2 * math.pi * R * C))`
- we get `(3.46+/-0.05)e+02`
- This value, aka 346Hz, is well within our measured $10^2 \pm \sqrt{10} \times 10^2$Hz
- using python
- ```
  FCalculated = 1/(2 * math.pi * R * C)
  FMeasured = ufloat(1e2, (10 ** 0.5) * 1e2)
  print(FMeasured)
  Sigma = (FCalculated.n - FMeasured.n) / (FCalculated + FMeasured).s
  print(Sigma)
  ```
- We get a measured sigma of 0.7777
- Since our sigma is <3, we can say the values corroborate eachother.

* 4

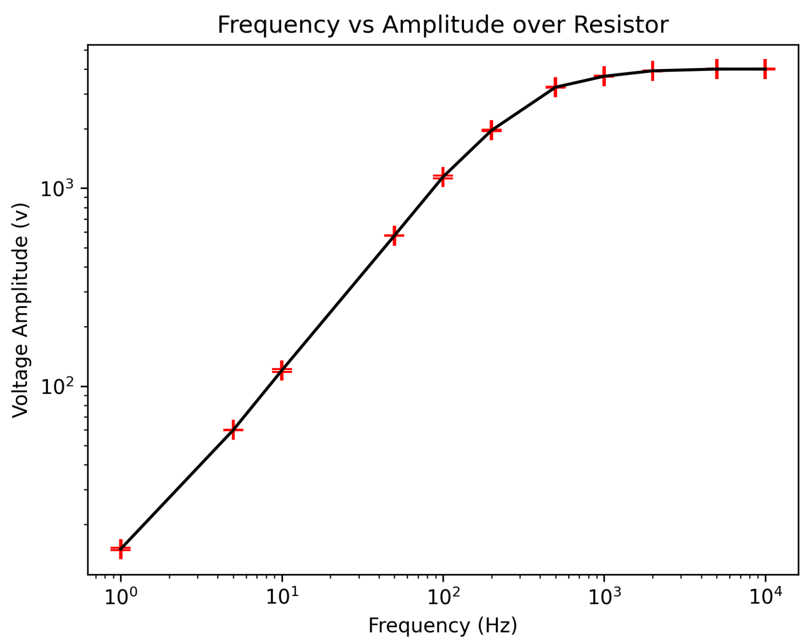

Figure 3: file:///home/burrs/Repositories/School/src/10week/physics/FreqAmplRes.png

  -
  ```

- Similarly to part 3, Frequency uncertainty is 1%, and voltage amplitude uncertainty is half of the lowest discrete jump. So if the oscilloscope measures 1140mV, the jump between each value is 20mV.
- this corroborates with our first graph. At higher frequencies, the capacitor has a voltage that converges to 0, which means a voltage amplitude of V, aka voltage in. and at lower frequencies the capacitor has a voltage amplitude converging to 0, since the capacitor will just have the voltage of the Vin.
- we can also visually see that the cutoff frequency is when the voltage sort of hits its ceiling.
- The name I can [[think]] of the circuit is just the name of the graph: 'Frequency vs Amplitude over Resistor'.
- If i were to think of a name that sort of describes the information/meaning of the results it would be somethine more on the lines of 'How frequency will cause the voltage of a capacitor to approach 0'.